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Chips connecting lives

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Riding on a time machine to 2041 is exciting yet intriguing, like the feeling of a child holding a kaleidoscope. Look through it; you will see the mesmerizing world of technologies delivering superfast connectivity and magical performance across millions of devices. It's a world where robots take control over the industrial process, where drones engage in rescue operations, where people track every item, process, or person in their network in real-time and takes immediate actions if anything goes wrong. It's where the country is equipped with automation systems to ensure the security of the people and the optimal use of resources. Zoom your camera to virtual reality and feel the wonders of the world right from your little room. Think of virtual shopping at your favorite brands or a stroll through your dreamland in a quest for a villa, all of these while enjoying a coffee at the comfort of your home. The possibilities are infinite.

The World Economic Forum has identified five technologies that are transforming the global production systems and unleashing a new wave of competition among producers and countries alike. They are the internet of things (IoT), Artificial Intelligence (AI), advanced robotics, wearables, and 3D printing. The growth of 5G has accelerated the adoption of these technologies across a wide range of industries.

In 2041, we will witness an era of devices powered by smart chips featuring the best-in-class technology thanks to the SoC innovation and R&D initiatives happening in the sector. The world of chips is expanding, from the traditional CPU chips to the complex robotic chips capable of performing multitudes of tasks at incredible speed. In a devices-driven world, chips will play a lead role in enabling communications, enhancing productivity, and improving lifestyle. From mobile phones to smart home solutions, and from robots to cloud computing systems, innovation in chip technology will transform the way products are designed, manufactured, assembled, distributed, consumed, serviced, discarded, and even reused. As technology evolves, chips will grow smaller and greener, to power smarter and better products with yet another set of incredible features.

However, future progress cannot be limited to technology advancements alone; it's also about how the benefits of the technology are imparted to the cross-section of the society impartially. The progress is to be measured in terms of the overall societal change brought by them, eliminating the digital divide, and thus contributing to the nation's overall digital transformation goals. Thus, MediaTek envisions 2041 where technology democratization is at its best, and with that goal, we work towards making our technology accessible to every individual in the country. In addition to this, MediaTek foresees a great future for Make in India initiatives in the semiconductor industry. India has already initiated the process of setting up commercial semiconductor wafer fab units, so we expect much growth in this line in the coming years. MediaTek has been leveraging the local talent and resources and has already associated with indigenous manufacturers across a wide range of emerging segments including AIoT, WiFi 6, smart home, edge AI, and more. We expect explosive growth across these segments, alongside the growth of 5G-enabled smartphones powered by incredible chipsets.

The pandemic has brought about a quantum leap in technology-enabled transformation across societies. Digital adoption has risen following the citizen-centric initiatives adopted by the government, as well as the shift in work/study patterns. This trend will evolve further in the post COVID era, based on the learning acquired during the crisis. Thus, enterprises and governments are required to adopt strategies to deal with the new requirements from the community, for example, trends like hybrid workplaces, which are gaining momentum in the West. Virtually every sector will be impacted by the dynamic market changes, so adopting the best technologies to address the need for speed, performance, security, and analytics will be the main differentiator.

Lastly, an outlook will be incomplete without the commitment to sustainable practices. The growing adoption of electronic devices is going to have a severe impact on the environment. Disposal and recycling of e-waste, as well as safe handling of the materials, will emerge as a challenge. Responsible e-waste management with active participation from governments and organizations will demand the necessary attention. Consumer awareness initiatives, legislation from government bodies, as well as the adoption of responsible corporate sustainability practices by industry participants will help drive a sustainable ecosystem. In chip manufacturing, where a large scale of natural resources is required, environmental impact can be high. The only way to reduce the impact is to optimize the resources and make the best use of the products once it is out, to the world.